Introduction

A cross-tabulation shows the joint distribution of two categorical variables. It is the first look at whether the variables are associated and is the input to chi-squared, Fisher's exact, and other tests of independence.

Prerequisites

The reader should know what a one-way frequency table is.

Theory

For two categorical variables X with r levels and Y with c levels, the cross-tabulation is an $r \times c$ table whose (i, j)-th cell holds the count of observations with $X = i$ and $Y = j$. Typically reported forms:

- **Joint counts**: raw n_{ij}.
- **Row percentages**: $100 \cdot n_{ij} / \sum_j n_{ij}$. Answers "given $X = i$, what is the distribution of Y?"
- **Column percentages**: $100 \cdot n_{ij} / \sum_i n_{ij}$. Answers "given $Y = j$, what is the distribution of X?"
- **Overall percentages**: $100 \cdot n_{ij} / n$. Joint probabilities.

Assumptions

Purely descriptive; no distributional assumption.

R Implementation

```r
library(dplyr)
library(janitor)
library(gtsummary)

set.seed(2026)
cohort <- tibble::tibble(
  sex   = factor(sample(c("Female", "Male"), 300, replace = TRUE)),
  arm   = factor(sample(c("Placebo", "Active"), 300, replace = TRUE)),
  event = factor(sample(c("Yes", "No"), 300, replace = TRUE, prob = c(0.25, 0.75)))
)

# Base R
table(cohort$arm, cohort$event)

# Percentages
addmargins(prop.table(table(cohort$arm, cohort$event), margin = 1), margin = 2)

# Tidy with janitor
cohort |>
  tabyl(arm, event) |>
  adorn_percentages("row") |>
  adorn_pct_formatting(digits = 1) |>
  adorn_ns()

# gtsummary
cohort |>
  tbl_cross(row = arm, col = event, percent = "row", margin = "column")

# Three-way via xtabs
xtabs(~ sex + arm + event, data = cohort)
```

Output & Results

A typical 2x2 table:

	Event: No	Event: Yes	Total
Placebo	118 (78.7%)	32 (21.3%)	150
Active	104 (69.3%)	46 (30.7%)	150

Row percentages show the event rate within each arm; column percentages would instead show what fraction of events came from each arm.

Interpretation

For association analysis, report counts and one of row, column, or joint percentages – the one that matches the causal direction of your research question. "Among placebo patients, 21.3% experienced an event" is a row-percentage statement appropriate when arm precedes event.

Practical Tips

- Always state whether percentages are row, column, or overall; it is a common source of reader confusion.
- Sparse cells (count < 5) motivate Fisher's exact test rather than chi-squared.
- For three-way or higher tables, consider log-linear models (`loglm` in `MASS`).
- `gtsummary::tbl_cross()` produces publication-ready output directly from a data frame.
- Report totals on the margins; they let readers check arithmetic quickly.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Descriptive statistics and Table 1